Kwame Nkrumah University of
Science & Technology, Kumasi, Ghana

ME 270
Manufacturing Technology

Lecture 1: Introduction

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Course Objectives

- Review the introduction to manufacturing.
- Explain the connection between processing and the mechanical and physical properties of materials.
- Explain the fundamentals of welding - Fusion (Oxy-fuel gas and electric arc welding processes) and solid-phase welding processes
- Explain the various material removal processes in manufacturing.
- Introduction to Computer Aided Manufacturing

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Reading Materials

- Manufacturing Engineering and Technology (7th edition) by Serope Kalpakjin and Steven Schmid April 11, 2013. ISBN-10: 01331287441 ISBN-13: 978- 0133128744
- Fundamentals of Modern Manufacturing: Materials, Processes and Systems (3rd edition) by Mikell P. Groover, 2007. ISBN – 10: 0-471-74485-9; ISBN – 13: 978-0-471-74485-6

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Grading

- 30% Continuous assessment
- 70% Exams

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Course Content

- Introduction to manufacturing
- Connection between processing and the mechanical and physical properties of materials
- Casting and moulding processes
- Fundamentals of welding
- Material removal processes
- Introduction to support functions in manufacturing
- Introduction to computer aided manufacturing

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Introduction to Manufacturing Technology

- To understand manufacturing technology, you must first understand manufacturing.
- Manufacturing is all about adding value to merchandise. This could be through the use of labour, machinery, chemicals, formulation, or biological processes to add value to a raw product before selling it. ***All activities involved in translating raw materials into finished product*** are part of the manufacturing process.

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Introduction to Manufacturing Technology

- Any technology that shapes or influences the manufacturing process is part and parcel of manufacturing technology.

This means that you can choose to be

- A biological technologist involved in developing biological manufacturing processes
- A ***robotics experts*** who designs and fine tunes automation robots on the production line
- A ***mechanical expert*** interested in the manufacturing equipment or the product itself
- Electronics engineer who powers the machines being used
- A chemist in chemical related production lines

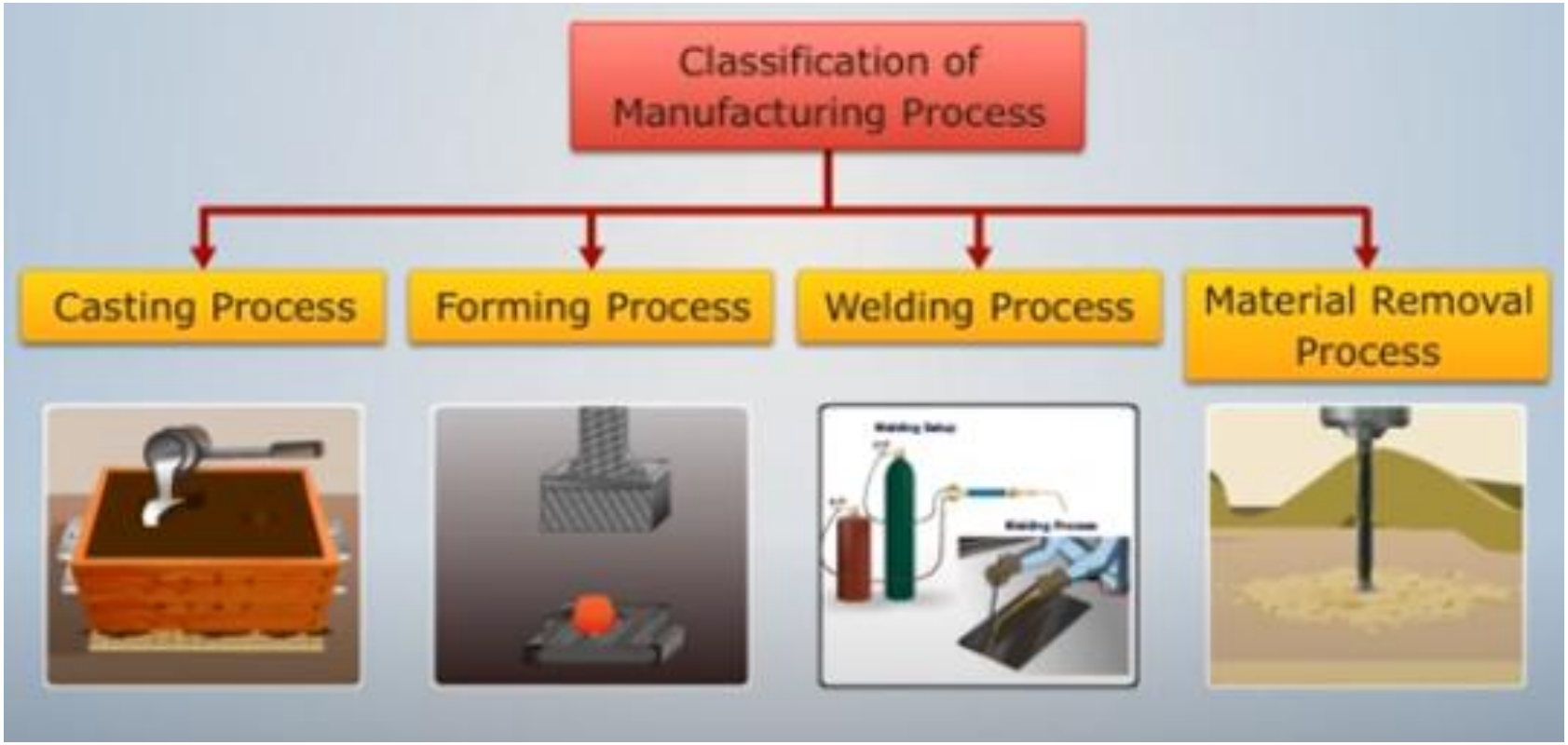
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Materials Science and Manufacturing Technology

- Material science basically deals with the property of materials and their behaviour under different circumstances and environments.
- The properties and behaviour of materials is extremely useful and necessary if those materials are to be worked on to manufacture any sort of finished products.
- Manufacturing is done by use of metals and materials of different kinds such as semiconductors and alloys.
- Knowledge of materials and their properties is, therefore, important in the field of manufacturing technology.

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Classification of Manufacturing Processes



Casting and Moulding Processes

- **Casting** is a manufacturing process in which a liquid material is usually poured into a mould, which contains a hollow cavity of the desired shape, and then allowed to solidify.
- The solidified part is also known as a casting, which is ejected or broken out of the mould to complete the process.

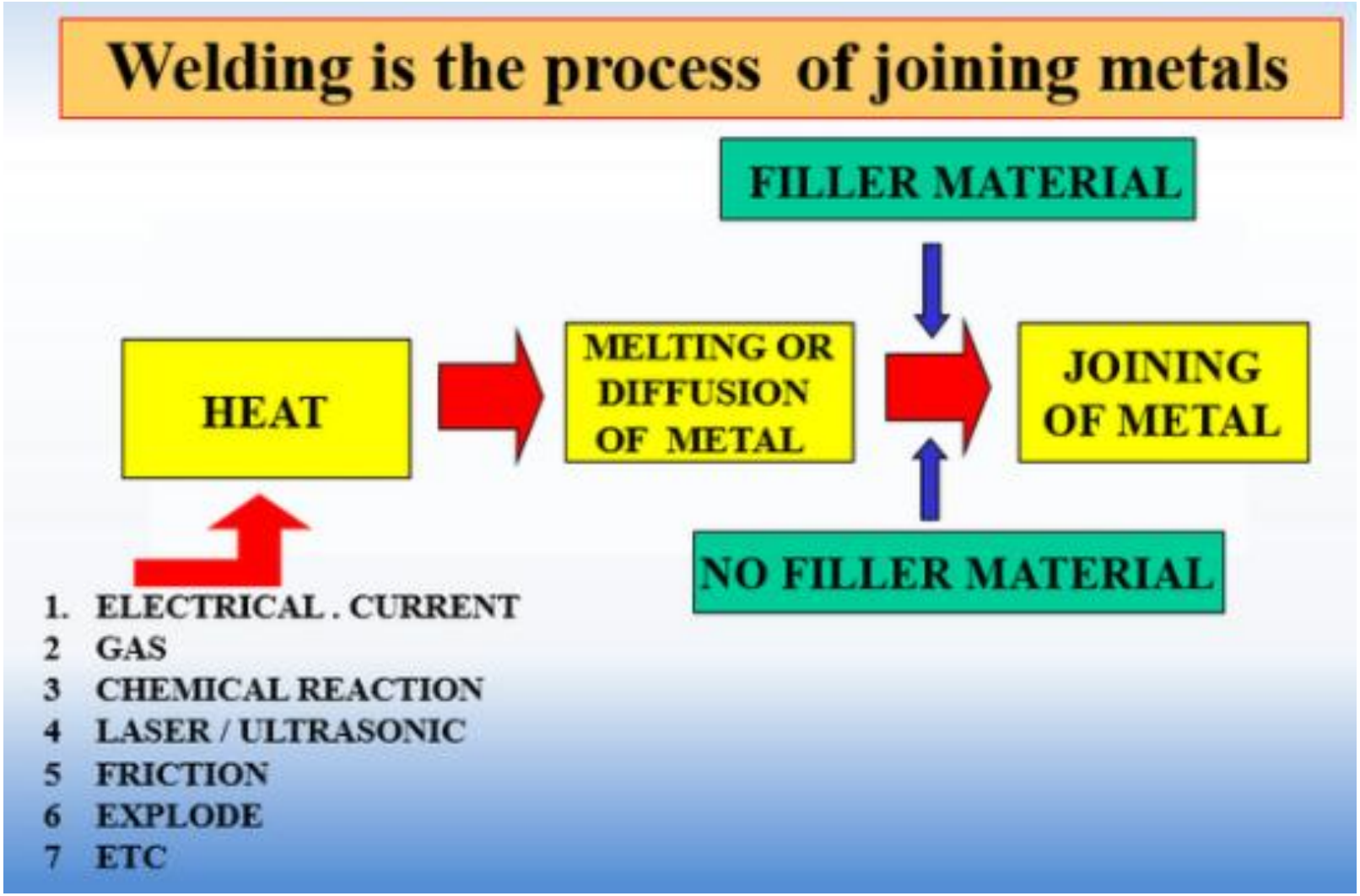


Casting and Moulding Processes

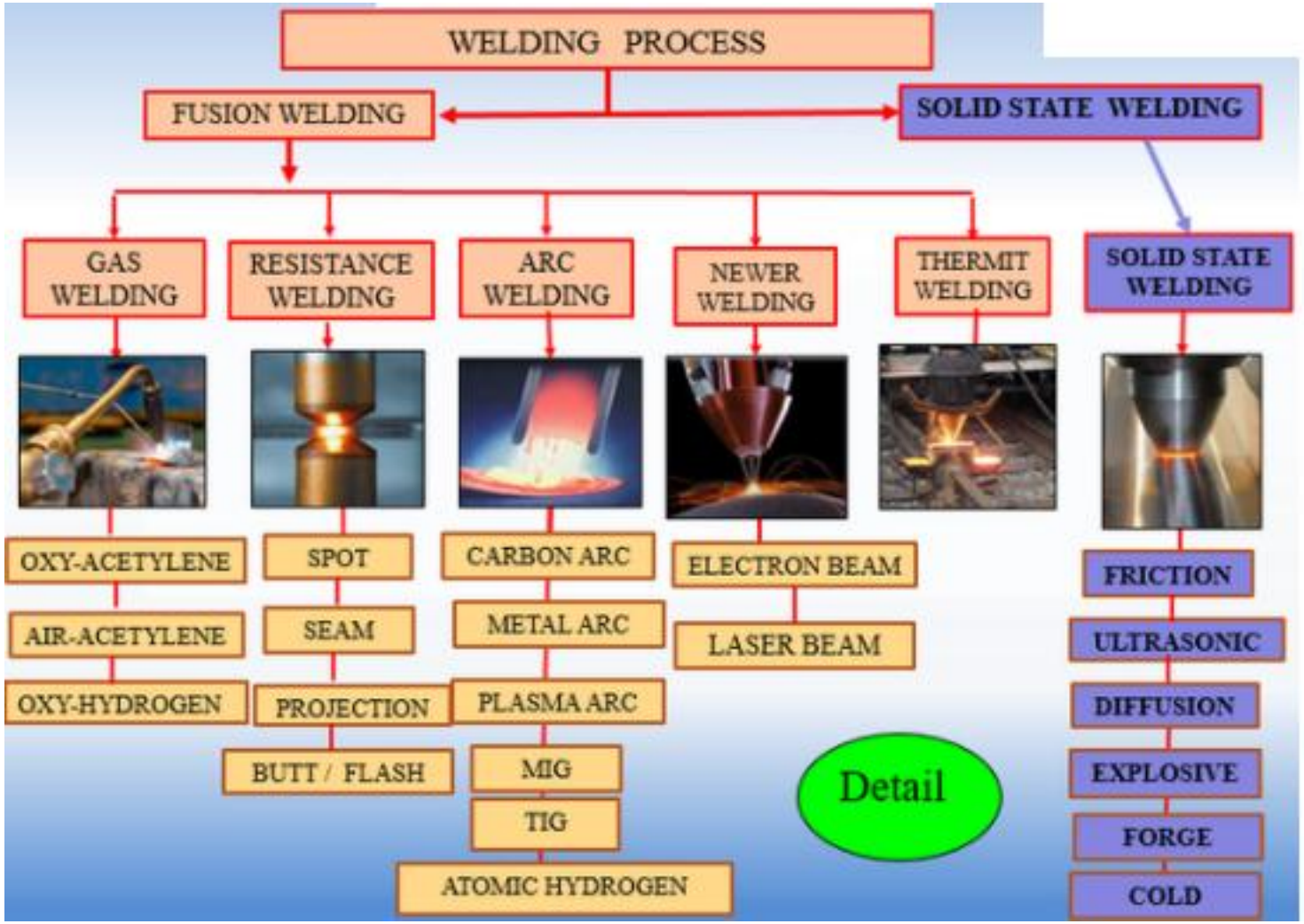
- **Moulding** is a technique through which a material, often plastic, but also metal, rubber, or powder mixtures is shaped on the outline of a die or mould. There are many different techniques for moulding materials, just as there are many different applications for each process.



Welding

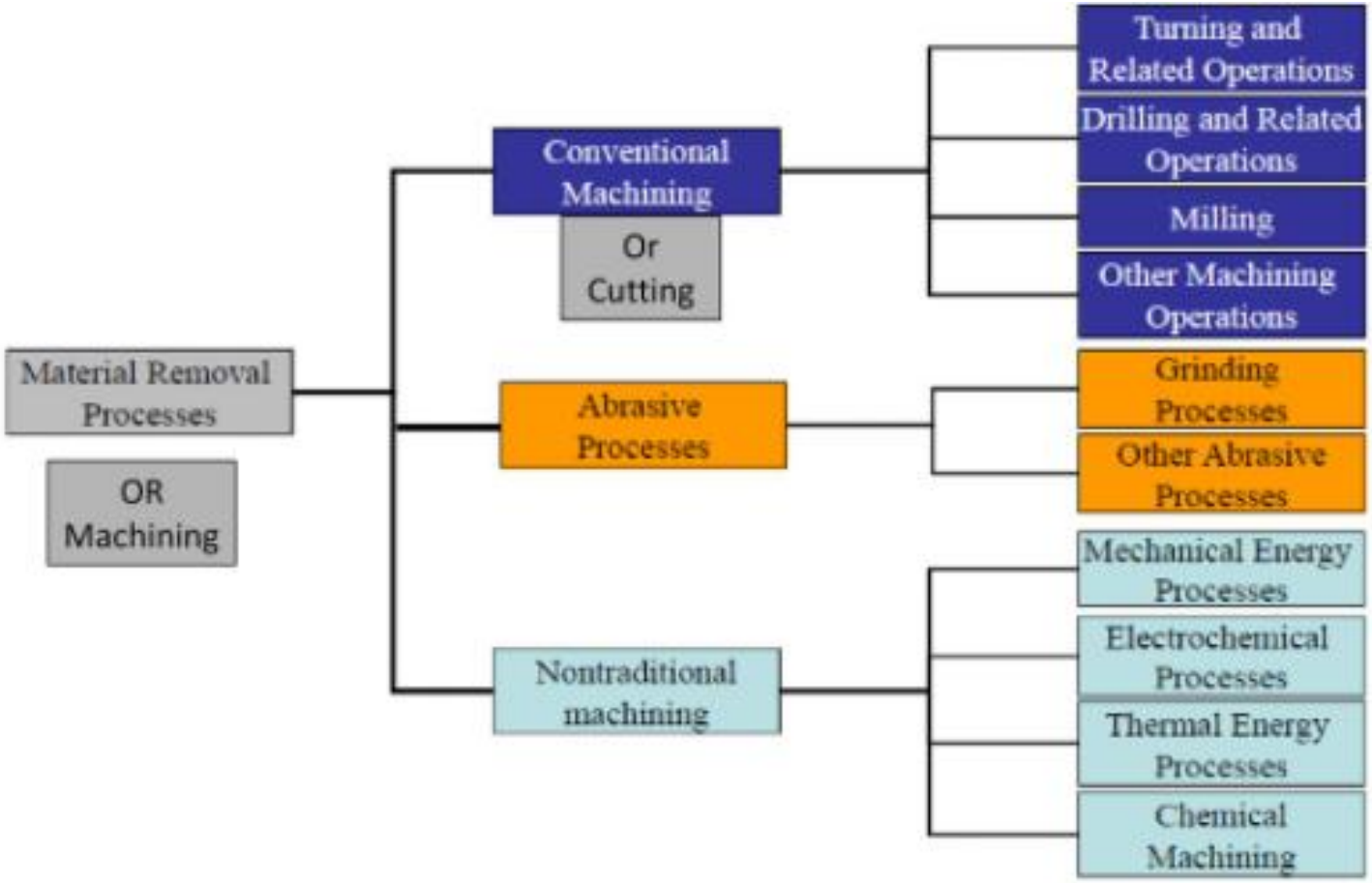

Welding is the process of joining metals
FILLER MATERIAL
HEAT
MELTING OR DIFFUSION OF METAL
JOINING OF METAL
NO FILLER MATERIAL
1. ELECTRICAL . CURRENT
2 GAS
3 CHEMICAL REACTION
4 LASER / ULTRASONIC
5 FRICTION
6 EXPLODE
7 ETC

Types of Welding



Material removal processes

- Material removal process is a type of manufacturing process in which the final product is obtained by removing excess metal from the stock.



Manufacturing Technologies Trends

- Computer Aided Manufacturing
- 3D Printing
- Robotics
- AI and Machine Learning
- Internet of Things (IoT)

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Thank You